PPP-B-35C
April 9, 1976
SUPERSEDING
Fed. Spec. PPP-B-35B
April 8, 1965

FEDERAL SPECIFICATION

BAGS, TEXTILE, SHIPPING, BURLAP, COTTON AND WATERPROOF LAMINATED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers shipping bags made from cotton cloth; burlap cloth; and textile laminated to paper, polyethylene, or other specified materials; for use in domestic and overseas shipment of supplies and materials.

1.2 Classification.

1.2.1 Types, classes, and grades. The bags covered by this specification shall be of the following types, classes, and grades as specified (see 6.2):

Type I - Standard burlap bag (see 3.3.1).
Type II - Standard cotton bag (see 3.3.2).
Type III - Laminated textile bags (see 3.3.3).

Class 1 - Paper - asphalt - textile.
Class 2 - Paper - asphalt - paper - asphalt - textile.
Class 3 - Paper - asphalt - textile - asphalt - paper.
Class 4 - Paper - asphalt - paper - adhesive - textile.
Class 5 - Paper - latex (or other laminant) - textile.
Class 6 - Polyethylene - adhesive - paper - adhesive - textile.
Class 7 - Other combination as specified by the procuring agency.

Grade I - Cotton sheeting.
Grade II - Cotton osnaburg.
Grade III - Jute burlap.

Type IV - Textile inner bag (see 3.4).

1.2.2 Styles. Bags shall be of the following styles as specified (see 6.2):

Style A - Sewn side and bottom seams (for types I, II, or III)(see 3.3.4).
Style B - Cemented longitudinal seams, sewn and taped bottoms (for type III only) (see 3.3.5).

FSC 8105

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

L-P-378	-	Plastic Sheet and Strip, Thin Gauge, Polyolefin.
QQ-S-781	-	Strapping, Steel, and Seals.
VV-S-190	-	Sealing Compound (Dip-Coat).

Federal Standards:

FED-STD-101	-	Preservation, Packaging, and Packing Materials: Test Procedures.
FED-STD-123	-	Marking for Shipment (Civil Agencies).
FED-STD-191	-	Textile Test Methods.
FED-STD-356	-	Commercial Packaging of Supplies and Equipment.
FED-STD-751	-	Stitches, Seams and Stitching.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-T-3530	-	Thread and Twine; Mildew Resistant or Water Repellent Treated.

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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US Department of Health, Education and Welfare

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

American Society for Testing and Materials (ASTM) Standards

D 917 - Bleeding Resistance of Asphalted Paper at Elevated Temperature.
D 1327 - Woven Burlap Fabrics Saturated with Bituminous Substances for Use in Waterproofing.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Technical Association of the Pulp and Paper Industry (TAPPI) Publications

T402 - Standard Conditioning and Testing Atmospheres for Paper, Board, Pulp Handsheets and Related Products.
T404 - Tensile Breaking Strength of Paper and Paperboard (Using Pendulum-Type Tester).
T410 - Weight Per Unit Area (Basis Weight of Substance) of Paper and Paperboard.
T414 - Internal Tearing Resistance of Paper.
T441 - Water Absorbtiveness of Sized (Nonbibulous) Paper and Paperboard (Cobb Test).
T456 - Wet Tensile Breaking Strength of Paper and Paperboard.
T457 - Stretch of Paper and Paperboard.

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(Applications for copies should be addressed to TAPPI Publications Order Department, One Dunwoody Park, Atlanta, GA 30341.)

Burlap Mill Grade Listing

(Applications for copies should be addressed to the Textile Bag Manufacturer's Association, 518 Davis Street, Evanston, Illinois.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Domestic and overseas use. Textile and textile laminated bags manufactured in conformance with this specification are for use in the domestic and overseas shipment (coastwise ocean shipment and shipments abroad) of the commodities listed in table XI, for commodities procured by the pound and table XII, for commodities procured by the bushel. The bag construction used for a specific commodity shall be not less than the construction specified for the bag number in tables I, II, or III as applicable. The commodities packed in textile inner bags as specified in table XIII shall be prepared for shipment in accordance with the appendix of this specification.

3.2 Materials. The materials used in the construction of bags shall be new and unsoiled, except that type I burlap bags may be furnished new or used as specified (see 6.2).

3.3 Construction. The bags provided for in this specification are those of standard construction commonly furnished by and known to the industry. Bags, except type I, shall be new and unused and furnished in standard construction.

3.3.1 Type I, standard burlap bag. Type I bags shall be fabricated from jute burlap conforming to table I for the applicable bag number. Seams shall be style A (see 3.3.4).

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TABLE I. Type I burlap bags for domestic and export shipments, minimum requirements of burlap cloth

Bag number	Burlap cloth 1/		
	Minimum width inclusive of selvage 1/	Weight per linear yard by 40 inch width ± 5 percent	Texture (ends and picks per inch ± 0.5)
	Inches	Ounces	
B1	36	7.5	8 x 9
B2	36	8	9 x 10
B3	36	10	11 x 12
B4 2/	36	10.4	12 x 13
B5	36	10.5	11 x 12
B6	36	12	11 x 12
B7	36	14	11 x 12
B8	40	7.5	8 x 9
B9	40	8	9 x 10
B10	40	10	11 x 12
B11	40	10.5	11 x 12
B12	40	12	11 x 12
B13	40	14	11 x 12
B14	45	7.5	8 x 9
B15	45	8	9 x 10
B16	45	10	11 x 12

1/ The breaking strength of the burlap shall be not less than burlap manufactured by a group C mill of the Burlap Mill Grade Listing (see 2.2).

2/ Special quality.

3.3.1.1 Type I, used bags. Type I, used bags shall be number 1, thoroughly reconditioned bags. Used bags shall be clean, dry, and free of holes, tears, frayed threads, objectionable odors, and foreign matter inpregnated in, or adhering to the sides of the bags. If mended, there shall be no more than two mends per bag and maximum size of each mend shall be not more than 2-inches. Bags with markings shall be reversed so that the outside does not contain markings. In all other respects used bags shall conform to the requirements of this specification. When procurement documents specify type I used bags, new type I standard burlap bags may be furnished provided the new bags meet all of the other requirements of the used bags.

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3.3.2 Type II, standard cotton bag. Type II bags shall be fabricated from cotton sheeting or osnaburg conforming to table II for the applicable bag number. Sheeting shall be unbleached. Seams shall be style A (see 3.3.4).

TABLE II. Type II cotton bags for domestic and export shipments, minimum requirements for cotton cloth

Bag number	Fabric	Width 1/	Linear yards per pound	Texture (ends and picks) per inch)	Breaking strength (grab method) Warp	Filling
		Inches			Pounds	Pounds
C21	Sheeting	36	2.85	48 x 48	65	55
C22	Sheeting	40	2.50	48 x 48	65	55
C23	Sheeting	40	2.85	48 x 48	60	50
C31	Sheeting	31	5.00	48 x 48	42	40
C32	Sheeting	37	4.00	48 x 44	45	40
C33	Sheeting	40	3.25	48 x 44	55	40
C34	Sheeting	40	3.75	48 x 44	48	35
C35	Sheeting	40	4.25	44 x 40	40	35
C41	Sheeting	27	6.80/7.25	44/48 x 44/48	31	31
C42	Sheeting	27	9.00/9.50	44/40 x 44/48	24	24
C42B	Sheeting	26 2/		42/44 x 36/40	24	24
C43	Sheeting	32	6.25	38/42 x 38/42	32.5	32.5
C43B	Sheeting	30 3/		40/44 x 36/38	32.5	32.5
C44	Sheeting	36	6.05/6.15	40 x 40	28	28
C46	Sheeting	40	3.60	56 x 56/60	45	40
C47	Print	38.5	5.35/5.50	64 x 56/60	25	22
C47	Muslin	36 4/		66 x 52/56	25	22
C51	Osnaburg	30	2.28/2.35	38/40 x 28	75	75
C52	Osnaburg	36	1.90/1.95	38/40 x 28	75	75
C53	Osnaburg	36	2.28/2.35	38/40 x 26	60	60
C54	Osnaburg	36	2.85/2.95	32/28 x 26	50	50
C55	Osnaburg	36	3.90/4.05	32/28 x 28/26	40	40
C56	Osnaburg	40	1.70/1.76	38/40 x 28	75	75
C57	Osnaburg	40	2.05/2.11	38/40 x 26	60	60
C58	Osnaburg	40	2.55/2.65	32 x 28/26	50	50
C59	Osnaburg	40	3.55/3.65	32/28 x 28/26	40	40
C70	Leno Cotton	32	5.35	7-1/4 x 8	30	30
C71	Woven Paper	32		6-1/2 x 6	35 26	30 (dry) 22 (wet)

1/ Pro rata widths to above construction will be acceptable.

2/ Regular bleach, made from 27 inch - 9.00/9.50-yard sheeting.

3/ Regular bleach, made from 32 inch - 6.25-yard sheeting.

4/ Made from 38-1/2 - 5.35/5.50-yard print.

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3.3.3 Type III, laminated textile bags. Type III bags shall be fabricated with the plies as specified, in the sequence from inside to the outside ply conforming to table III for the applicable bag number. Seams shall be style A or B as specified (see 3.3.4, 3.3.5, and 6.2).

TABLE III. Type III, laminated textile bags for export and special commodity and storage conditions, minimum requirements for ply construction (see 6.1.1)

Bag No.	Type III laminated textile bag ply construction, sequence inside to outside 1/					
	Class 1					
	Paper 4/		Asphalt 4/		Textile	
	Pounds per ream		Pounds per ream		Grade	
P1	30		80		I	- 2.85 yd.
P2	30		80		I	- 2.50 yd.
P3	30		80		II	- 2.35 yd.
P4	30		80		II	- 2.11 yd.
P5	30		80		III	- 7-1/2 oz.
P6	30		80		III	- 10 oz.
P7	30		80		III	- 10.5 oz.
P8	30		80		III	- 12 oz.
P9	30		80		III	- 14 oz.
	Class 2					
	Paper 4/	Asphalt 4/	Paper 4/	Asphalt 4/	Textile	
	Pounds per ream	Pounds per ream	Pounds per ream	Pounds per ream	Grade	
P10	30	50	30	80	I	- 2.85 yd.
P11	30	50	30	80	I	- 2.50 yd.
P12	30	50	30	80	II	- 2.35 yd.
P13	30	50	30	80	II	- 2.11 yd.
P14	30	50	30	80	III	- 7-1/2 oz.
P15	30	50	30	80	III	- 10 oz.
P16	30	50	30	80	III	- 10.5 oz.
P17	30	50	30	80	III	- 12 oz.
P18	30	50	30	80	III	- 14 oz.

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TABLE III. Type III, laminated textile bags for export and special commodity and storage conditions, minimum requirements for ply construction (see 6.1.1) (cont'd)

Bag No.	Type III laminated textile bag ply construction, sequence inside to outside 1/				
	Class 3				
	Paper 4/	Asphalt 4/	Textile	Asphalt 4/	Paper 4/
	Pounds per ream	Pounds per ream	Grade	Pounds per ream	Pounds per ream
P20	30	70	I - 2.85 yd.	70	30
P21	30	70	I - 2.50 yd.	70	30
P22	30	70	II - 2.35 yd.	70	30
P23	30	70	II - 2.11 yd.	70	30
P24	30	70	III - 7.5 oz.	70	30
P25	30	70	III - 10 oz.	70	30
P26	30	70	III - 10.5 oz.	70	30
P27	30	70	III - 12 oz.	70	30
P28	30	70	III - 14 oz.	70	30
	Class 4				
	Paper 4/	Asphalt 4/	Paper 4/	Latex or other adhesive 2/	Textile
	Pounds per ream	Pounds per ream	Pounds per ream		Grade
P30	30	70	30	I	- 2.85 yd.
P31	30	70	30	I	- 2.50 yd.
P32	30	70	30	II	- 2.35 yd.
P33	30	70	30	II	- 2.11 yd.
P34	30	70	30	III	- 7.5 oz.
P35	30	70	30	III	- 10 oz.
P36	30	70	30	III	- 10.5 oz.
P37	30	70	30	III	- 12 oz.
P38	30	70	30	III	- 14 oz.
	Class 5				
	Paper before creping 4/		Latex or other adhesive 2/		Textile
	Pounds per ream				Grade
P40	30			I	- 2.85 yd.
P41	30			I	- 2.50 yd.
P42	30			II	- 2.35 yd.
P43	30			II	- 2.11 yd.
P44	30			III	- 7.5 oz.
P45	30			III	- 10 oz.
P46	30			III	- 10.5 oz.
P47	30			III	- 12 oz.
P48	30			III	- 14 oz.

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TABLE III. Type III, laminated textile bags for export and special commodity and storage conditions, minimum requirements for ply construction (see 6.1.1) (cont'd)

Bag No.	Type III laminated textile bag ply construction, sequence inside to outside 1/			
	Class 6			
	Polyethylene Adhesive 2/	Paper before creping 4/	Adhesive 2/	Textile
		Pounds per ream		Grade
P50		30		1 - 2.85 yd.
P51		30		1 - 2.50 yd.
P52		30		II - 2.35 yd.
P53		30		II - 2.11 yd.
P54		30		III - 7.5 oz.
P55		30		III - 10 oz.
P56		30		III - 10.5 oz.
P57		30		III - 12 oz.
P58		30		III - 14 oz.

1/ When paper is specified, it shall be plain kraft paper conforming to table V. When wet-strength paper is required, it shall conform to table VI. Textiles for laminated bags shall conform to table IV for the grade specified.

2/ Latex, vegetable glues, or other types of adhesive shall be as specified by the procuring agency (see 6.2).

3/ Polyethylene shall be of the thickness specified in 3.3.3.3.

4/ A ream weight tolerance of ± 5 percent is permitted.

3.3.3.1 Textiles. Laminated shipping bags shall be fabricated in part from the following grades of textiles, as specified (see 6.2).

Grade I - Cotton sheeting
Grade II - Cotton osnaburg
Grade III - Jute burlap

The grades of textiles shall conform to the requirements shown in table IV.

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TABLE IV. Grade of textiles for type III laminated textile bags, minimum requirements

Grade	Fabric	Minimum width inclusive of selvage	Linear yards per pound ± 5 percent	Ounces per linear yard ± 5 percent	Texture (ends and picks per inch) ± 0.5	Breaking strength (grab method)	
						Warp	Filling
		Inches				Pounds	Pounds
I	Sheeting	36	2.85		48x48	65	55
I	Sheeting	40	2.50		48x48	65	55
II	Osnaburg	30	2.35		38/40x26	75	75
II	Osnaburg	36	2.35		38/40x26	60	60
II	Osnaburg	40	2.11		38/40x26	60	60
III	Burlap	36		7.5	8x9		
III	Burlap	40		7.5	8x9		
III	Burlap	36		10	11x12		
III	Burlap	40		10	11x12		
III	Burlap	36		10.4	12x13	Not less than	
III	Burlap	36		10.5	11x12	cloth manufac-	
III	Burlap	40		10.5	11x12	tured by group C	
III	Burlap	36		12	11x12	mills (see 2.2)	
III	Burlap	40		12	11x12		
III	Burlap	40		14	11x12		

3.3.3.2 Paper. Laminated shipping bags shall be fabricated in part from kraft paper, plain, or when specified (see 6.2) of plain kraft and wet-strength kraft paper. The paper shall be stretchable in one, or both directions, as specified (see 6.2). The paper shall have a stretch of not less than 17 percent in two directions before laminating, when tested as specified in 4.2.1.1.2.

3.3.3.2.1 Plain kraft paper. Plain kraft paper shall be 100 percent unbleached kraft fiber, resin sized and, before creping, shall conform to the requirements of table V. When used for the packing of food, the materials used in the production of kraft paper shall conform with the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

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TABLE V. Plain kraft paper, before creping 1/

Ream basis weight 24 x 36-500	Minimum average tearing strength		Minimum average tensile strength per inch wide	
	Machine direction	Total machine direction plus cross direction	Cross direction	Total cross direction plus machine direction
Pounds	Grams	Grams	Pounds	Pounds
30	40	90	11	32
40	70	150	15	40

1/ A tolerance of ± 5 percent will be permitted for the value shown for ream weight.

3.3.3.2.2 Wet-strength kraft paper and markings. Wet-strength kraft paper shall be 100 percent unbleached kraft fiber, resin sized and, before creping, shall conform to the requirements of table VI. The wet-strength paper shall be distinctly colored, stained, printed, or marked for identification purposes with longitudinal red stripes, spaced not less than 2 inches nor more than 10 inches apart, across the paper width and each stripe shall be not less than 1/8 inch in width. No other grade of paper used in textile and paper laminated shipping bags shall be striped as described above. When wet-strength paper is used as the outer ply of the bag, the identification shall appear on the external surface.

TABLE VI. Wet strength kraft paper 1/

Ream 1/ basis wt. 24x36 500	Minimum average bursting strength	Minimum average tearing strength			Minimum average tensile strength per inch width	Minimum average wet tensile strength per inch width	
		Machine direction	Total machine direction plus cross direction	Cross direction	Total cross direction plus machine direction	Cross direction	Total cross direction plus machine direction
Pounds	Pounds	Grams	Grams	Pounds	Pounds	Pounds	Pounds
30	27	35	80	12	36	3.0	9.0
40	36	50	120	16	45	4.0	11.5

1/ A ream weight tolerance of ± 5 percent is permitted.

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3.3.3.2.3 Variations in paper requirements. Variations below minimum specification requirements for cross tensile strength of plain and wet-strength kraft paper components will be permitted provided they are compensated by an increase in the machine direction tear in ratio of 5 units of tear to 1 unit of tensile; similar variations in combined tensile strength will be permitted provided they are compensated by increase in the combined tear in the same ratio. Similarly, variations in machine direction tearing strength shall be permitted provided the variations are compensated by an increase in the cross direction dry tensile strength in the ratio of 1 unit of dry tensile strength to 5 units of tearing strength; and variations in combined tearing strength shall be permitted provided the variations are compensated by an increase in the combined dry tensile strength in the same ratio.

3.3.3.2.4 Values shown in tables V and VI. All values shown in tables V and VI are for paper conditioned to equilibrium in accordance with TAPPI Standard T 402, Conditioning Paper and Paperboard for Testing.

3.3.3.3 Polyethylene. Polyethylene used for type III, class 6 bags shall have a minimum thickness of 0.002 inch, when tested as specified in 4.3.2.4.

3.3.3.4 Bonding agent. Bonding agents used in bag construction for military overseas shipment shall not fail when tested as specified in 4.3.2.1.

3.3.3.4.1 Asphalt. When an asphalt compounded bonding agent is used, the softening point shall be not less than 140°F., when tested as specified in 4.3.2.6. Asphalt shall be applied in uniform and continuous film. The asphalt laminated sheet shall show no visual bleeding or cracking when tested as specified in 4.3.2.2. The weight of asphalt per ply of laminating material shall average not less than 70 pounds per ream, 24 by 36 - 500 sheets when tested as specified in 4.3.2.7.

3.3.4 Style A (sewn side and bottom seams). Unless otherwise specified (see 6.2), all bags shall be style A (open mouth, see 3.3.7) and shall be manufactured with sewn side and bottom seams sufficiently strong to develop not less than 80 percent of the strength of the material when tested as specified in 4.2.1.1.1.

3.3.4.1 Stitching. The minimum average number of stitches shall be 3.0 to the inch, machine sewn and of stitch type 301 or 401 of FED-STD-751.

3.3.4.2 Seams for type I bags (burlap). Selvaged edges shall be sewn with flat seams, and raw edges with single turnover seams (see fig. 1). Raw top edges of bags shall be hemmed.

3.3.4.3 Seams for type II bags (cotton). Flat seams shall be used for type II bags with the stitching not less than 1/2 inch from edge of the cloth (see fig. 1). Raw top bags shall be hemmed.

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3.3.4.4 Seams for type III bags (laminated textile). Single turnover seams not less than 1/2 inch wide shall be used (see fig. 1).

3.3.5 Style B (type III bags only). Style B bags, type III only, shall be manufactured with cemented longitudinal seams and sewn and taped bottom seam (open mouth, see 3.3.7).

3.3.5.1 Cemented longitudinal seams. Cemented longitudinal seams for style B bags shall have a minimum 1 inch overlap and shall be cemented with a bonding agent which will effect a bond equivalent in strength to the material of the bag when tested as specified in 4.3.2.1.

3.3.5.2 Sewn and taped bottom seams. Bottoms seams shall be sewn with a flat seam with stitching as specified in 3.3.4.1 not less than 1/2 inch from edge of cloth. The tape specified in 3.3.5.2.1 shall be cemented over the stitches so that it will extend equally over and across the faces of the bag ± 1/8 inch, and protrude not less than 1/2 inch nor more than 2 inches beyond both edges of the bag.

3.3.5.2.1 Tape. The tape shall be not less than 2-1/2 inches wide and shall be made from class 1 material of this specification or from material offering equivalent properties.

3.3.6 Waterproofing of seams (type III bag). When specified (see 6.2), all sewn seams of type III bags shall be dipped in a sealing compound specified in 3.3.6.1 to a depth at least 1/4 inch beyond the upper edge of the seam on the bag body.

3.3.6.1 Sealing compound. The compound for waterproofing and sealing the seams of bags shall be as specified in VV-S-190.

3.3.7 Bag top. Unless otherwise specified, (see 6.2) the bag top shall have an open-mouth top. When a valve top is specified for type I, II, or IV bags (tuck-in sleeve or inner sleeve), it shall be factory sewn (see 6.2).

3.3.8 Size (capacity). The capacity of the sacks shall be such as to contain the amount and kind of material specified in the contract or order (see 6.2).

3.3.9 Odor or flavor. Materials used in the construction of sacks for subsistence items shall not impart a foreign odor to the contents (see 4.3.2.5).

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3.3.10 Printing or stenciling, and marking. All shipping bags shall be legibly printed or stenciled with the marking specified in the contract or order. The ink used shall be black, nonfading, and water resistant.

3.3.11 Bag manufacturer's identification. Each bag shall be marked on one side only with the specification number, the bag manufacturer's name and address and the bag type, and when applicable, class and grade. All markings shall be placed in a conspicuous position not less than 5-1/2 inches from the bottom of the bag in black letters not less than 5/16 inch high, except that "PPP-B-35" shall be in letters approximately 3/4 inch high. Arrangement of markings shall be as follows:

(Example only):

COMPLIES WITH
FED. SPEC. PPP-B-35
JOHN DOE BAG CO.
CHICAGO, ILLINOIS
BAG TYPE _____
BAG NO. _____

This requirement does not apply to type I, used bags, and to new type I bags when substituted for used type I bags.

3.3.11.1 Type III bags. Type III bags, for overseas shipment, shall have the words "FOR OCEAN SHIPMENT" positioned immediately below the bag manufacturers identification in letters 1-inch to 1-1/2 inches high.

3.3.12 Bag manufacturer's certificate. When required (see 6.2), certificate shall be in the form, size, and wording as provided in Rule 40 of the National Motor Freight Classification or Uniform Freight Classification.

3.3.13 Mildewproofing. When specified (see 6.2), all material used in bag construction shall be mildewproofed in accordance with MIL-T-3530, type 1, class 1.

3.4 Type IV, textile inner bags. Textile inner bags shall be fabricated from material designated in table XIII. The material shall conform to the physical characteristics of table VII. Materials which are acceptable substitutes for 31 inch - 5.00 yard cotton sheeting are listed in table VIII, and materials which are acceptable substitutes for 31 inch - 5.00 yard or 36 inch - 2.35 cotton osnaburg are listed in table IX. The construction shown in these tables are minimum requirements, and materials in other widths of equivalent weight, construction, breaking strength, and type will be acceptable. Unless otherwise specified, all cotton cloths are unbleached (see 6.2).

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TABLE VII. Textile material for textile inner bags, minimum requirements

Fabric	Width	Linear yards per pound	Texture (ends and picks per inch)	Breaking strength (warp and filling) (grab method)
	Inches			Pounds
Cotton sheeting, unbleached calendered	32	6.25	38/42 x 38/42	32.5
Regular bleach cotton 1/	30	6.20/6.25	35/36 x 35/36	32.5
Cotton sheeting (see table VIII)	31	5.00	48 x 48	41
Cotton sheeting (see table IX)	36	2.85	48 x 48	60
Cotton sheeting (see table IX)	36	2.35	38/40 x 26/26	60
Cotton print	38.5	6.25	60 x 48	25
Cotton print	38.5	5.50	64 x 56	25
Cotton print	45	4.75	64 X 56	31.8
Cotton print	32	6.62/6.25	64 x 56	25
Burlap	40	1.60	11 x 12	2/

1/ Regular bleach is made from cotton grey goods woven in 31.5/32 inch widths, bleached and filled. All other cotton cloths are unbleached.

2/ Not less than cloth manufactured by group "C" mills (see 2.2).

TABLE VIII. Textile material for textile inner bags, acceptable substitute for 31 inch - 5.00-yard cotton sheeting, minimum requirements

Fabric	Width	Linear yards per pound	Texture (ends and picks per inch)	Breaking strength (warp and filling) (grab method)
	Inches			Pounds
Cotton sheeting	40	2.85	48 x 48	55
Cotton sheeting	37	4.00	48 x 44	43
Cotton sheeting	40	3.25	48 x 44	48
Cotton sheeting	40	3.75	48 x 40	43
Cotton osnaburg 1/	36	2.95	32/28 x 26	50

1/ Acceptable for coarse products such as beans, peas, pearl barley, etc.

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TABLE IX. Textile material for textile inner bags, acceptable substitutes for 31 inch-5.00 yard or 36 inch-2.85 yard cotton sheeting or 36 inch-2.35 yard cotton osnaburg, minimum requirements

Fabric	Width	Linear yards per pound	Texture (ends and picks per inch)	Breaking strength (warp and filling) (grab method)
	Inches			Pounds
Cotton sheeting	40	2.50	48 x 48	60
Cotton osnaburg	30	2.35	38/40 x 26	75

3.4.1 Seams. Textile inner bags shall be manufactured with seams sufficiently strong so that when tension is applied across the bag seam, the cloth shall fail before failure of the seam. Flat seams may be used on the cotton bag provided that the stitching is not less than 1/2-inch from the edge of the material. Export seams shall be used on the burlap bags when securing cut edges.

3.4.2 Size.

3.4.2.1 Five- and ten-pound inner bags. The dimension of the 5-pound and 10-pound inner bags shall be such that after closure they contain their respective quantities without being tightly filled. When filled and closed, inner bags shall be of such shape that they can be tightly packed into the outer shipping bag.

3.4.2.2 Fifty-pound inner bag. The dimensions of the 50-pound inner bag shall be such that when filled and closed, it shall contain 50 pounds of the product without being tightly filled and, when on its face, it shall have a uniform overall thickness of not more than 30 percent of its width, and the ratio of length to width of between 1.5 and 2 to 1.

3.4.3 Printing. No printing is required on textile inner bag for 50 pounds of the product. The cotton bag for 5 or 10 pounds of the product shall be printed or legibly stenciled with nontoxic ink to indicate the name of the product and the quantity contained therein.

3.5 Liners. When specified in the contract or order (see 6.2), liners shall be furnished and shall conform to the minimum requirements of table X and with 3.5.1 and 3.5.2, as applicable. Seams shall be sewed and cemented in accordance with accepted commercial practice. Size of the liner shall be a minimum of 3 inches longer than the bag in which liner to to be inserted.

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3.5.1 Paper liners. Paper liners shall be made from 100 percent unbleached kraft fiber.

3.5.2 Polyethylene liner. Polyethylene used for liners shall be not less than 0.002 inch thick, and shall conform to L-P-378.

TABLE X. Liners, waxed, polyethylene, and asphalt laminated paper, minimum requirements

Liner	Construction						
	Impregnation		Lamination			Stretch	
	Paper 1/ Pounds per ream	Wax Percent	Paper 1/ Pounds per ream	Ashpalt Pounds per ream	Paper 1/ Pounds per ream	One direction Percent	Both directions Percent
A 2/	45	15				30	
B 2/	30	15				30	30
C			30	30	30	30	
D			30	30	30	30	30
E	Polyethylene, thickness, .002 inches.						

1/ Basis weight per ream (24 by 36-500) before crinkling, creping or processing.

2/ Liners A and B are interchangeable.

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TABLE XI. Commodities packed in textile or laminated shippings bags, minimum acceptable bag requirements 7/

Commodity	Civilian				Military		Extra
	Domestic 1/		Overseas		Overseas		
	Capacity (pounds)		Capacity (pounds)		Capacity (pounds)		
	50	100	50	100	50	100	
Aluminum fluoride		P42A P45A		P3A/B P6A/B	P3A/B P6A/B	P3A/B P6A/B	None
Aluminum hydrate		P42A P45A		P3A/B P6A/B	P6A/B P6A/B	P3A/B P6A/B	None
Aluminum oxide		P42A P45A		P3A/B P6A/B	P3A/B P6A/B	P3A/B P6A/B	None
Aluminum stearate		P42A P45A		P3A/B P6A/B	P3A/B P6A/B	P3A/B P6A/B	None
Aluminum sulfate		B10 P42A P45A		P3A/B P6A/B		P3A/B P6A/B	None
Ammonium chloride		P22B P25B		P22B P25B		P22A/B P25A/B	Note 2/
Ammonium phosphate		P12A P15A		B12 P12B P15B		P12A/B P15A/B	None
Ammonium sulfate		B10 P3A P5A		B10 P6A/B P3A/B		P3A/B P6A/B	None
Antimony oxide		P42A/B P44A/B		P45A/B P5A/B		P45A/B P5A/B	None
Barium chloride		P3A P5A		P6A/B		P3A/B P6A/B	
Barium chromate	P24B	P24B	P25B	P25B	P22A/B P25A/B		Note 2/
Barium nitrate	P25B	P25B	P25B	P25B	P22A/B P25A/B	P22A/B P25A/B	Note 2/
Barytes	P3A	B10 C51 P3A P5A	P6A/B	P6A/B	P3A/B P6A/B	P3A/B P6A/B	None

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TABLE XI. Commodities packed in textile or laminated shipping bags, minimum acceptable bag requirements 7/ (cont'd)

Commodity	Civilian				Military		Extra
	Domestic 1/		Overseas		Overseas		
	Capacity (pounds)		Capacity (pounds)		Capacity (pounds)		
	50	100	50	100	50	100	
Bentonite	P44A	P44A	P45A/B	P45A/B		P45A/B	None
Blanc fixe		P3A P5A		P6A/B		P3A/B P6A/B	None
Borax		C32 B3 P44A		P45A/B P45A/B		P12A P15A/B P45A/B	None
Boric acid		B6 P3A/B P5A/B		B6		P12A/B P15A/B	None
Calcium carbonate (commercial ground limestone)		P44A		P45A/B	P3A/B P6A/B P45A/B	P3A/B P6A/B P45A/B	None
Calcium carbonate, fine (precipitated) (carbonate of lime)		P5A		P6A/B		P3A/B P6A/B	None
Calcium chloride		P34B		P35B		P35B/A	Note 3/
Calcium chromate		P14A/B		P15A/B	P22A/B P25A/B		Note 2/
Calcium hydroxide (hydrated lime)	P44A/B P5A/B	P44A/B P5A/B	P45A/B P6A/B	P45A/B P6A/B	P25A/B	P1A/B P6A/B P45A/B	None
Calcium oxide (75-80 lbs.)		P14B		P15B		P12B/A	Note 3/
Calcium phosphate		P14A/B		P15B		P12A/B P15A/B	None
Calcium silicate (silicate of lime)		P44A P5A		P45A P46A		P45A P46A	None
Calsomine		P3A P5A		P6A/B	P3A/B P6A/B		None

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TABLE XI. Commodities packed in textile or laminated shipping bags, minimum acceptable bag requirements 7/ (cont'd)

Commodity	Civilian				Military		Extra
	Domestic 1/		Overseas		Overseas		
	Capacity (pounds)		Capacity (pounds)		Capacity (pounds)		
	50	100	50	100	50	100	
Casein, dry		P44A P5A		P45A P6A		P3A/B P6A/B P45A/B	None
Cement, portland or high early 94#		C51 B7 B4		P15B	No requirement		None
Cement, refractory 94#	P3A/B P5A/B P15A/B	C53 P3A/B P5A/B P15A/B B7 C51	P15A/B P6A/B P12A/B	P15A/B P6A/B P12A/B	P6A/B P12A/B P15A/B	P6A/B P12A/B P15A/B	None
Citric acid		P3A/B P5A/B		P6A/B		P3A/B P6A/B	None
Clay, fire and refractory	P3A/B P5A/B P15A/B	C21 C31 B10 P3A/B P5A/B P15A/B	P15A/B P12A/B	P15A/B P12A/B P6A/B		P3A/B P6A/B P12A/B P15A/B	None
Clay, kaoline and air floated		C31 B10		P45A P5A		P3A P5A P45A	None
Clay, other than air floated		C31 B10	P45A P5A	P45A P5A		P3A P45A	None
Copper sulfate (Penta)		C56 B10 P15A/B		P15A/B		P11A/B P15A/B	Note 4/
Corn cob meal		P44A		P45A		P45A P3A P6A	None

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TABLE XI. Commodities packed in textile or laminated shipping bags, minimum acceptable bag requirements 7/ (cont'd)

Commodity	Civilian				Military		Extra
	Domestic 1/		Overseas		Overseas		
	Capacity (pounds)		Capacity (pounds)		Capacity (pounds)		
	50	100	50	100	50	100	
Cotton seed meal		C32 B10 P44A		C54 B10 P45A		P45A P3A P6A.	None
DDT, 100 percent	P5EB	P5EB	P6EB	P6EB		P3EB P6EB	None
Diatomaceous earth		B12 P44A P5A		P45A P6A	P3A/B P6A/B P45A/B	P3A/B P6A/B P45A/B	None
Disodium phosphate		P5A		P6A		P1A/B P6A/B	None
Dog food, dry		C34 P44A		P45A		P42A P45A	None
Feed, mineral (animal or poultry)	C51 C31 B3	C54 C32 B10 P5A P6A		P6A/B		P3A/B P6A/B P45A/B	None
Feldspar		P44A/B P45A/B		P45A/B P6A/B		P3A/B P6A/B P45A/B	None
Fertilizer, mixed 80/100 pounds		P4A/B B10 B9 C55 C32 C54		P6A/B P15A/B B10 C53		P3A/B P6A/B P15A/B	None
200 pounds		C57		P6A/B P15A/B			

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TABLE XI. Commodities packed in textile or laminated shipping bags, minimum acceptable bag requirements 7/ (cont'd)

Commodity	Civilian				Military		Extra
	Domestic 1/		Overseas		Overseas		
	Capacity (pounds)		Capacity (pounds)		Capacity (pounds)		
	50	100	50	100	50	100	
Fullers earth		B10 P44A P5A	P45A	B10 P45A P6A		P3A P6A P45A	None
Fullers earth, granular		B10 P44A P5A		B10 P45A P6A		P3A P6A P45A	None
Glues and pastes, dry		B10 P44A P5A		P6A/B		P3A/B P6A/B	None
Graphite	P24B		P24B	B13 P24B	P22B P25B	P22B P24B	Outer ply shall be W/C kraft
Grog. (insulating) Granular calcined		P45A P6A		P45A P6A		P3A P6A P45A	None
Gypsum	P44A P5A	B44A P5A B10	P45A P6A	P45A P6A		P13A P6A P45A	None
Iron oxide	P5A	P5A	P6A/B	P6A/B	P3A/B P6A/B	P3A/B P6A/B	None
Lead formate	P5A	P5A	P6A/B	P6A/B	P3A/B P6A/B	P3A/B P6A/B	None
Lead oxide	P5A	P5A	P6A/B	P6A/B	P3A/B P6A/B		None
Lead stearate	P5A	P5A	P6A/B	P6A/B	P3A/B P6A/B	P3A/B P6A/B	None
Lead sulfate	P5B	P5B	P5B P15B	P5B P15B	P3B P6B P15B	P3B P6B P15B	None

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TABLE XI. Commodities packed in textile or laminated shipping bags, minimum acceptable bag requirements 7/ (cont'd)

Commodity	Civilian				Military		Extra
	Domestic 1/		Overseas		Overseas		
	Capacity (pounds)		Capacity (pounds)		Capacity (pounds)		
	50	100	50	100	50	100	
Lead, dry, white		P6B		P15B		P12B P15B	None
Lignin pitch	P25B	P25B	P25B	P25B		P22B P25N	Note 2/
Litharge, dry	P45B P6B	P45B P6B	P45B P6B	P45B P6B		P3B P6B P45B	None
Lithopone	P5A P6A	P5A P6A	P6A/B	P6A/B		P12A/B P15A/B	None
Magnesite, dead burned	P44A P5A	P44A P5A	P45A P6A	P45A P6A	P3A P6A P45A	P3A P6A P45	None
Magnesium oxide (commercial magnesium quicklime-see calcium oxide quicklime)		P15B		P15B		P12A/B P15A/B	None
Magnesium silicate	P5A	P5A	P6A/B	P6A/B	P3A/B P6A/B	P3A/B P6A/B	None
Magnesium stearate	P44A P5A	P44A P5A	P6A/B P45A/B	P6A/B P45A/B	P3A/B P6A/B P45A/B	P3A/B P6A/B P45A/B	None
Magnesium sulfate (epsom salts)		P15B		P15B		P12B/A P15B/A	Note 3/
Mica		P45A P5A		B6 P45A P6A		P3A P6A P45A	None
Mortar cement	P3A/B P6A/B P15A/B	P3A/B P6A/B P15A/B	P6A/B P15A/B P12A/B	P6A/B P15A/B P12A/B	P3A/B P6A/B P12A/B P15A/B	P3A/B P6A/B P12A/B P15A/B	None

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TABLE XI. Commodities packed in textile or laminated shipping bags, minimum acceptable bag requirements 7/ (cont'd)

Commodity	Civilian				Military		Extra
	Domestic 1/		Overseas		Overseas		
	Capacity (pounds)		Capacity (pounds)		Capacity (pounds)		
	50	100	50	100	50	100	
Mortar, refractory (heat setting)	P3F/B P6A/B P15A/B	P3A/B P6A/B P15A/B	P6A/B P15A/B P12A/B	P6A/B P15A/B P12A/B	P3A/B P6A/B P12A/B P15A/B	P3A/B P6A/B P12A/B P15A/B	None
Nickel oxide		P15B		P15B		P12B P15B	None
Ochre		P5B P44B		P6B P45B		P3B P6B P45B	None
Plaster, building	P5A P44A	P5A P44A	P6A P45A	P6A P45A	P3A/B P6A/B P45A/B	P3A/B P6A/B P45A/B	None
Plaster, magnesium	P5A P44A	P5A P44A	P6A P45A	P6A P45A	P3A P6A P45A	P3A P6A P45A	None
Potassium aluminum sulfate	P14B	P14B	P15B	P15B		P12B P15B	None
Potassium bicarbonate	P14A/B P15A/B	P14A/B P15A/B	P15A/B	P15A/B		P22A/B P25A/B	Note 2/
Potassium bichromate	P14B P15B	P14B P15B	P15B	P15B		P12B P15B	None
Potassium carbonate (commercial potash)	P6A/B	P6A/B	P6A/B	P6A/B		P22A/B P25A/B	Note 2/
Potassium chloride	P14B P15B	P14B P15B	P15B	P15B		P12B P15B	None
Potassium chromate	P14B P15B	P14B P15B	P15B	P15B		P12B P15B	None

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TABLE XI. Commodities packed in textile or laminated shipping bags, minimum acceptable bag requirements 7/ (cont'd)

Commodity	Civilian				Military		Extra
	Domestic 1/		Overseas		Overseas		
	Capacity (pounds)		Capacity (pounds)		Capacity (pounds)		
	50	100	50	100	50	100	
Potassium magnesium sulfate	P14B P15B	P14B P15B	P15B	P15B		P12B P15B	None
Potassium nitrate	P15A/B	P15A/B	P15B	P15B		P12B P15B	None
Potassium sulfate	P15A/B	P15A/B	P15B	P15B		P12B P15B	None
Pumice	P3A P44A P45A	P3A P44A P45A B3	P45A/B P6A/B	P45A/B P6A/B	P3A/B P6A/B P45A/B	P3A/B P6A/B P45A/B	None
Rockwood, loose 40 pounds	P3A P44A		P45A P6A		P3A P6A P45A		None
Sand, dry		C51 P45A P6A		P6A/B P15A/B	No requirement		None
Seeds, grass, vegetable	C21 P44A P45A P5A	C21 C54 B9 P44A P45A P5A	C53	C53 B10 P3A/B P6A/B	No requirement		
Shells, oyster, crushed		C54 C21 B3		C53 C21 B3	No requirement		
Silica, pulverized (for molding and work)		C51 P45A P6A		P6A/B P15A/B		P12A/B P15A/B	

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TABLE XI. Commodities packed in textile or laminated shipping bags, minimum acceptable bag requirements 7/ (cont'd)

Commodity	Civilian				Military		Extra
	Domestic 1/		Overseas		Overseas		
	Capacity (pounds)		Capacity (pounds)		Capacity (pounds)		
	50	100	50	100	50	100	
Soap, pure, powdered, granular and flake or chips	C31 P5A P44A P45A	C33K C34K C51 C56 P5A P44A P45A B8	C31 P45A/B P6A/B	C22 C21 C23 C57 P45A/B P6A/B B10K B3 B8K	P3A/B P6A/B P45A/B		Note 3/
Soda ash, granular (dense)		P45A P6A		P45A P6A		P3B/A P6B/A P45B/A	Note 3/
Soda ash (light)		P45A P6A		P45A P6A		P3B/A P6B/A P45B/A	Note 3/
Soda ash (briquettes)	P45A P6A		P45A P6A			P3A/B P6A/B P45A/B	None
Sodium bicarbonate	P14A/B P15A/B	B3 P14A/B P15A/B	P15A/B	C21 P15A/B		P12A/B P15A/B	None
Sodium bichromate	P14A/B P15A/B	P14A/B P15A/B	P15B	P15B		P11B P15B	None
Sodium chloride (rock salt)	P45A	C32 B3 B45A	P6A	P6A		P12B/A P15B/A	None
Sodium chloride (fine)	P45A	C21 B3 B45A	P6A	P6A	P3B/A P3B/A		Note 5/

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TABLE XI. Commodities packed in textile or laminated shipping bags, minimum acceptable bag requirements 7/ (cont'd)

Commodity	Civilian				Military		Extra
	Domestic 1/		Overseas		Overseas		
	Capacity (pounds)		Capacity (pounds)		Capacity (pounds)		
	50	100	50	100	50	100	
Sodium meta silicate	P5A/B P6A/B	P5A/B P6A/B	P6A/B P15A/B	P6A/B P15A/B		P12A/B P15A/B	None
Sodium nitrate		B10 B6A		P15B		P12B/A P15B/A	Note 3/
Sodium phosphate	P5A P6A	P5A P6A	P6A/B P15A/B	P6A/B P15A/B		P3A/B P15A/B P6A/B	None
Sodium resinate	P5A P6A	P5A P6A	P6A/B P15A/B	P6A/B P15A/B	P3A/B P12A/B P15A/B P6A/B	P3A/B P12A/B P15A/B P6A/B	None
Sodium silicates	P5A P6A	P5A P6A	P6A/B P15A/B	P6A/B P15A/B	P3A/B P6A/B P12A/B P15A/B	P3A/B P6A/B P12A/B P15A/B	None
Sodium silicofluoride	P15B	P15B	P15B	P15B	P12B P15B	P12B P15B	None
Sodium sulfate (anhydrous)	P15B	P15B	P15B	P15B		P12B/A P15B/A	Note 3/
Sodium sulfate (Glaubers salt)	P15B	P15B C53	P15B	P15B		P3A/B P6A/B	None
Sodium sulfite	P5A/B P6A/B	P5A/B P6A/B	P6A/B	P6A/B		P12A/B P15A/B	None
Sodium thiosulfate	P5A/B P6A/B	P5A/B P6A/B	P6A/B	P6A/B		P10B/A P15B/A	Note 3/
Starch, laundry	P44A P45A	P44A P45A C54	P45A	P45A	P42A P45A	P42A P45A	None

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TABLE XI. Commodities packed in textile or laminated shipping bags, minimum acceptable bag requirements 7/ (cont'd)

Commodity	Civilian				Military		Extra
	Domestic 1/		Overseas		Overseas		
	Capacity (pounds)		Capacity (pounds)		Capacity (pounds)		
	50	100	50	100	50	100	
Starch, powdered 200 pounds 100 pounds	 P44A P45A	C57 C58 P44A P45A	 P45A	 P45A	P42A P45A	P42A P45A	None
Stearic acid, (flake or powdered) 150 pounds	 P44A P45A	B3 P44A P45A	 P45A	B3 P45A		P3A/B P6A/B	None
Sulfur, refined, ground		C33 C21 P45A B8 B1	 P45A	C21 C53 B10K	P12A/B P15A/B	P12A/B P15A/B	None
Superphosphate, granular		B3 P6A		P6A		P3B/A P6B/A	Note 6/
Superphosphate, pulverized		C54 B3 P6A		P6A		P3B/A P6B/A	Note 6/
Superphosphate, triple		P6A/B		P6A/B P15A/B		P3B/A P6B/A	Note 6/
Talc	P5A P44A P45A	C32 P5A P44A P45A	P45A P6A	P45A P6A	P3A/B P6A/B	P3A/B P6A/B	None
Tetrasodium purophosphate	P5A/B P6A/B	P5A/B P6A/B	P6A/B P15A/B	P6A/B P15A/B		P3B/A P6B/A	Note 3/
Thermit	P6A/B	P6A/B	P6A/B P15A/B	P6A/B P15A/B	P3A/B P6A/B P12A/B P15A/B	P3A/B P6A/B P12A/B P15A/B	None

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TABLE XI. Commodities packed in textile or laminated shipping bags, minimum acceptable bag requirements 7/ (cont'd)

Commodity	Civilian				Military		Extra
	Domestic 1/		Overseas		Overseas		
	Capacity (pounds)		Capacity (pounds)		Capacity (pounds)		
	50	100	50	100	50	100	
Titanium dioxide	P4A	P4A	P13A	P13A		P13A	None
Tripoli powder (tripolite)	P5A P44A P45A	P5A P44A P45A	P45A P6A	P45A P6A		P3A P6A P42A P45A	None
Trisodium phosphate	P5A P6A	P5A P6A	P6A	P6A		P1B/A P6B/A	Note 3/
Wax, chipped, slab or crude, also 200 #		B3 P44A P45A		B3 P45A		P3A/B P6A/B	None
Zinc borate	P5A P6A	P5A P6A	P6A	P6A	P3A/B P6A/B	P3A/B P6A/B	None
Zinc chromate	P5A P44A P45A	P5A P44A P45A	P45A P6A	P45A P6A	P3A/B P6A/B	P3A/B P6A/B	None
Zinc oxide	P5A P44A P45A	P5A P44A P45A	P45A P6A	P45A P6A		P3A/B P6A/B P42A/B P45A/B	
Baler bags (also 200 pounds)	P5A P6A	P5A P6A	P6A/B P15A/B	P6A/B P15A/B	P3A/B P6A/B P12A/B P15A/B	P3A/B P6A/B P12A/B P15A/B	Note 3/

1/ The bag requirements for the civilian domestic shipment of a commodity shall be acceptable for the military domestic shipment of the same commodity.

2/ The outer ply of the bag shall be wet-strength kraft paper.

3/ The sewn seams shall be waterproofed (see 3.3.6).

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4/ The stitching shall be effected with oiled or waxed thread and the inner sheet of the bag shall be wax coated.

5/ Fifty pounds of the product shall be packed into a type C21 or type C31 bag which shall be overpacked in the bag designated and the sewn seams shall be waterproofed (see 3.3.6).

6/ The stitching shall be effected with acid-resistant thread and the sewn seams shall be waterproofed (see 3.3.6).

7/ Legend - B1 to B16 see table I; C21 to C71 see table II; P1 to P58 see table III: A/B or B/A indicates style of bag and is listed in order of preference; the letters "E" and "K" where they appear require a polyethylene and loose kraft paper liner, respectively.

3.6 Closures and markings. Filled bags furnished under this specification shall be closed, packaged, and packed and marked in accordance with the procedures outlined in the appendix.

3.7 Workmanship. Bags and liners shall be accurately and well made and free from defects which may affect serviceability, and shall conform to the minimum quality assurance provisions of section 4 of this specification.

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TABLE XII. Commodities packed in textile and laminated textile shipping bags, minimum acceptable bag requirements 1/

Commodity	Weight per bushel (pounds)	Textile bags										Laminated textile bags			
		Domestic cotton			Export cotton			Domestic burlap		Export burlap		Domestic shipment		Export shipment or hazardous storage 100- and 200-pound bag	
		Capacity (pounds)		Liner	Capacity (pounds)		Liner	Capacity (pounds)	Liner	Capacity (pounds)	Liner	Bag No.	Seam style	Bag No.	Seam style
		50	100		50	100		100		100					
Barley	48	C31	C54	-	C31	C53	-	B3	-	B3	-	-	-	P45-6	A
Beans-dried	60	C31	C54	-	C31	C52	-	B4/6	-	B4/6	-	-	-	P6-15	A/B
Bran-shorts	20	-	C34		-	C53	-	B8	-	B10	-	-	-	P45-6	A
Coffee bean roasted		-	C33	-	-	C52		B8		B10		-		P14-15-6	A/B
Corn-cracked		C31	C34		C51	C21		B3		B3		-		P5	A
Shelled	56	-	C54		-	C53		B3		B3		-		P5	A
Feed-mill	25	-	C34		-	C33		B8		B10		P44	A	P5	A
Mixed	V.	C31	C34		-	C23		B8		B10		P44	A	P45-6	A
Fertilizer	V.	-	C54		-	C53		B9		B10		P5	A/B	P6	A
Flour		C31	C32		C31	C21/53		B3		B3	B	P5	A	P15-3	A
Grits		-	C33	B	-	C23	B	B3	B	B10	B	P44	A	P6	A
Hominy		-	C34		-	C23		B8		B10		P44	A	P6	A
Meal-alfalfa		-	C33		-	C22		B8		B10	B	P44	A	P6	A
Bone			C33		-	C22/57		B9		B10	B	P44	A	P6	A
Corn	48	C31	C32		C31	C21/53		B2		B3	B	P44	A	P6	A
Cottonseed	33	-	C32		-	C21/53		B8		B10		P44	A	P6	A
Fish		-	C32	B	-	C21	B	B8		B10	B	P44	A	P6	A
Glandular		-	C33		-	C23		B8		B10		P44	A	P6	A
Linseed	56	-	C32		-	C21		B8		B10		P44	A	P6	A
Oat	32	C31	C33		C31	C23		B10		B10	B	P44	A	P6	A
Soya	52	-	C32		-	C22		B2		B10		P44	A	P6-55	A
Malt	38	-	C34		-	C23		B8		B10		P6	B	P6-15	B
Milo	50	-	C33		-	C22		B10		B10		P4	A	P45	A

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TABLE XII. Commodities packed in textile and laminated textile shipping bags, minimum acceptable bag requirements 1/ (cont'd)

Commodity	Weight per bushel (pounds)	Textile bags										Laminated textile bags			
		Domestic cotton			Export cotton			Domestic burlap		Export burlap		Domestic shipment		Export shipment or hazardous storage 100- and 200-pound bag	
		Capacity (pounds)		Liner	Capacity (pounds)		Liner	Capacity (pounds)	Liner	Capacity (pounds)	Liner	Bag No.	Seam style	Bag No.	Seam style
		50	100		50	100		100		100					
Milk-dry	-	-	-		-	-		-		-		-		P6-15-PE	A/B
Powdered	-	-	-		-	-		-		-		P54	PE/A	P6-15-PE	A/B
Oats	32	-	C33/34		-	C22/23		B14/8		B8		P44	A	P45	A
Onions(NB)	57	C71	C71		C71	C71		B8		B8		-		-	-
Peanuts-															
Shelled	60	-	C54		-	C21		B3		B3		P44		P45-15	A/B
Whole	22	C32	-		C23	-		B9		B9		P44		P45-15	A/B
Peas	60	C31	C54		C31	C52		B4/6		B4/6		P44		P45-6	A
Potatoes	60	C31	C34/55		C21	C53		B8/1		B3/1		-		-	-
Rice	44	C31	C54/21		C54	C53		B4/3		B4/6		P45		P6-45	A
Rye	56	-	C58		-	C57		B2		B3		P45		P6-45	A
Salt	50	C31	C33/21		C51	C54/21	B	B3		B3		P45	A	P6	A
Seeds-grass)															
Vegetable)	V.	C21	C21/54		C53	C53		B9		B10		P45-5		P6-3	A/B
Shells-Oyster		-	-		-	-		B8		B10		-		-	-
Crushed	60	-	C54/21			C53/21		B3		B3		-		-	-
Soap-Chips		C31	C34		C31	C23/57		B8		B10/3		P5		P6	A
Powdered	C	C31	C33/34	B	C31	C22/21	B	B8	B	B10	B	P5	P	P6	A
Starch			C33/34		-	C22/21		B10		B10		P45		P6-45	A
Sugar		C31	C21		C31	C21		B3	C44	B3	C44	P45		P6	A
Sulphur		-	C33/21		-	C21/53		B8/1	B	B10	B	P45		P45	A
Wax		-	C54	C44	-	C53	C44	B10	C44	B10	C44	P44		P45	A
Wheat	60	-	C21/53		-	C21/53		B3		B3/6		P45		P6	A

1/ Legend - B1 to B16 see table I; C21 to C71 see table II; P1 to P58 see table III; A/B or B/A indicates style of bag and is listed in order of preference; the letters "E" and "K" where they appear require a polyethylene and loose kraft paper liner, respectively.

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2-50# Bags to be inserted in 100# Baler: 36" -7 oz. burlap or 36" -2.95 yd. Osnaburg is domestic; 36"-10 oz. burlap or 36" -2.35 yd. Osnaburg if export: In P3 or P6 if waterproof. (NB) - Onion shipments standard open mesh paper or leno cotton bag.

TABLE XIII. Commodities packed in textile inner bags, minimum bag requirements

Commodity	Capacity per bag Pounds	Bag requirements
Flour, salt, beans, peas, or rice	5 or 10	31-inch-5.00 yard cotton sheeting
Sugar	5 or 10	38-1/2-inch-5.50 yard cotton print cloth or 45-inch - 4.75 yard cotton print cloth or 31-inch - 5.00 yard cotton sheeting
Flour	50	32-inch-6.25 yard cotton sheeting calandered or 30-inch-6.20 to 6.25 yard regular bleach 1/ or 31-inch-5.00 yard cotton sheeting
Salt or sugar	50	36-inch-2.85 yard cotton sheeting or 36-inch-2.35 yard cotton osnaburg
Beans, peas, rice or pearl barley	50	36-inch-2.85 yard cotton sheeting or 36-inch-2.35 yard cotton osnaburg or 36-inch/40-inch 10 ounce burlap

1/ Regular bleach is made from cotton grey goods, woven in widths 31-1/2 to 32 inches, bleached and filled, having 6.20 to 6.25 yards per pound. All other cotton cloths are unbleached.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated herein.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.1.1 Testing of textile materials. Textile materials described in this specification shall be tested in accordance with the applicable test methods of FED-STD-191 and as indicated below. The sample unit and the sample size shall be the specimen and the number of specimens, respectively, as indicated in the applicable test method. Failure of the average of the test results on all sample units to meet the specified requirements for the characteristic being tested shall be cause for rejection of the component or the involved end item.

Tests	Method
Width	5020
Weight	5041
Texture	5050
Breaking strength	5100

4.2.1.2 Testing of paper materials. The paper specified in this specification shall be tested for the applicable characteristics as indicated in table XIV, for each type and weight of material used. The sample unit shall be one piece full width of the roll by forty-eight inches long when purchase is on the roll basis, or two sheets full size when purchase is in sheet form. No more than one sample unit shall be drawn from any one roll or package of sheets. Five sample units randomly selected shall be tested with no evidence of failure to meet the specified requirements.

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INSTRUCTIONS FOR TESTING PAPER

TABLE XIV

CHARACTERISTIC	Specification Reference		Requirements Applicable To		Number Determinations Per Unit	Results Reported As		Inspect Level	AQL
	Requirement	Test Method		Lot Aver		Pass or Fail	Numerically to Nearest		
Basis weight:									
Plain kraft	Table V	4.3.1	-	X	2		0.1 pound		
Wet-strength kraft	Table VI	4.3.1	-	X	2		0.1 pound		
Tearing resistance 1/	Tables V & VI	4.3.1							
Machine direction			-	X	Avg. of 5		gram		
Cross direction			-	X	Avg. of 5		gram		
Tensile strength 1/	Tables V & VI	4.3.1							
Machine direction			-	X	Avg. of 5		0.1 lb/in width		
Cross direction			-	X	Avg. of 5		0.1 lb/in width		
Stretch	3.3.3.2	4.3.1		X	Avg. of 3		0.5 percent		
Sizing (resin)	3.3.3.2.1 and 3.3.3.2.2	4.3.1	X	-		X			
			X	-		X			

1/ See 3.3.3.2.3 for permissible variations.

2/ If failure is indicated, report description of failure.

3/ Test reports shall include all values on which results are based.

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4.2.2 End item inspection. The end items shall be examined for defects classified under 4.2.2.1, 4.2.2.2 and 4.2.2.3. The inspection levels and acceptable quality levels (AQL's) expressed as defects per hundred units shall be as shown in 4.2.2.4. The inspection lot shall be expressed in units of bags for examination under 4.2.2.1 and 4.2.2.2 and in units of bales or shipping containers for examination in 4.2.2.3.

4.2.2.1 Visual examination. The bags shall be examined for the defects classified in table XV. The sample unit for this examination shall be one bag and the liner when applicable.

TABLE XV. Examination for defects in material, construction, appearance and workmanship

Examine	Defect	Classification Major	Minor
Material	Not as specified.	151	
Construction and appearance	Not new material, except type I bag when used burlap is specified.	152	
	Any hole, tear, or frayed threads.	153	
	Not "open-mouth" top construction, unless otherwise specified.	154	
	Valve top bag, when specified (types I, II or IV - tuck-in sleeve or inner sleeve) not factory sewn.	155	
Type I bag	More than two mends per bag.	156	
Type I bag	Mend greater than 2 inches.	157	
Stitching	Not lock or chain stitch.	158	
	Less than 3.0 stitches per inch.	159	
Seams:			
Type I bag	Selvaged edge not sewn with flat seams: bottom or raw edges or raw-top bag not sewn with single turn-over seam.	160	
Type II bag	Not sewn with flat seams; raw-top bag not hemmed.	161	
	Stitching not 1/2 inch from edge of cloth.		201
Type III bag (style B)	Not cemented longitudinal seams: bottom seam not sewn and taped; not sewn with flat seam 1/2 inch from edge of cloth	162	

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TABLE XV. Examination for defects in material, construction, appearance and workmanship (cont'd)

Examine	Defect	Classification Major	Classification Minor
Seams: (cont'd) Type III bag (style B) (cont'd)	Cemented longitudinal seam less than 1 inch overlap		202
	Tape less than 2-1/2 inches wide.	163	
	Tape does not extend equally over and across the face of the bag ± 1/8 inch.		203
	Tape protrudes less than 1/2 inch or more than 2 inches beyond one or both edges of the bag.		204
	Sewn seams not uniformly and adequately dipped in sealing compound to a depth of not less than 1/4 inch beyond the upper edge of the seam.	164	
Lamination: Type III bag	Plies not in sequence from inside to outside as specified (see table III).	165	
	Paper not creped in one direction, or two directions, or creped in one direction and corrugated or pleated in the other direction.	166	
	Wet strength kraft paper not distinctly colored, stained, printed or marked with longitudinal red strips as specified (see 3.3.3.2.2).		205
Seams: Type IV bag	Not sewn as specified (see 3.4.1); stitching less than 1/2 inch from edge of cloth.	167	
Liner (when applicable)	Not laminated with material as specified; plies not in sequence as specified (see table X).	168	
Workmanship	Bag assembly not uniform.		206
	Stitching not uniform and complete over entire area of seam.		207
	Lamination (type III bag) not uniform; any bulge or crease in lamination.		208
	Not clean.		209

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TABLE XV. Examination for defects in material, construction, appearance and workmanship (cont'd)

Examine	Defect	Classification Major	Minor
Identification	Omitted, incomplete, incorrect, illegible; or improper size, location sequence, or method of application (see 3.3.11, 3.3.11.1 and 3.3.12).		210

4.2.2.2 Examination for dimensional defects. The bags shall be examined for defects in table XVI. The sample unit for this examination shall be one bag and liner when applicable.

TABLE XVI. Examination for dimensional defects

Examine	Defect	Classification Major
Bag Overall dimensions	Length or width not within ± 5 percent of specified dimensions.	151
Liner (when applicable)	Does not fit bag uniformly.	152
	Less than 3 inches longer than bag.	153

4.2.2.3 Packaging inspection. An examination shall be made to determine that packing and marking comply with the requirements of section 5. The sample unit shall be one shipping container fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects. Defects shall be as listed in table XVII.

TABLE XVII. Examination of packaging

Examine	Defect	Classification Major
Contents	Number of bags per container less than specified or as indicated.	151
	More than one size or kind of bag per container.	152
Packing	Not level specified; not in accordance with contract requirements.	153

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TABLE XVII. Examination of packaging (cont'd)

Examine	Defect	Classification Major
Packing (cont'd)	Overwrapping, as applicable, missing or not as specified.	154
	Closures, as applicable, not accomplished by specified methods or materials.	155
	Strapping or cords, as applicable, loose or not number required.	156
Marking	Incomplete, missing, or not as specified.	157
Weight	Net weight exceeds requirements.	158
Condition	Torn or otherwise damaged container which may result in loss or injury to contents.	159

4.2.2.4 Inspection levels and acceptable quality levels (AQLs) for examination. The inspection levels for determining sample size and acceptable quality levels (AQLs) expressed in defects per 100 units shall be as follows:

Examination paragraph	Inspection levels	AQLs Major	AQLs Total
4.2.2.1	S-4	2.5	6.5
4.2.2.2	S-2	2.5	
4.2.2.3	S-4	4.0	

NOTE: When the end item consists of bags with liners, the inspection levels and AQLs shall apply to the bag and liner individually and separately.

4.2.3 Testing of the end item (type III bags only). Type III bags shall be tested for the applicable characteristics as indicated in table XVIII for each lot presented for examination of one class and size only. The sample unit shall be one bag. Five sample units, randomly selected throughout the lot, shall be tested with no evidence of failure to meet the unit or lot average requirements, as applicable.

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INSTRUCTIONS FOR TESTING END ITEM

TABLE XVIII

CHARACTERISTIC	Specification Reference		Requirements Applicable To		Number Determinations Per Unit	Results Reported As		Inspect Level	AQL
	Requirement	Test Method	Individ Unit	Lot Aver		Pass or Fail 1/	Numerically to Nearest 2/		
Bonding agent	3.3.3.4	4.3.2.1	X		2	X			
Asphalt:									
Bleeding	3.3.3.4.1	4.3.2.2	X		1	X			
Cracking	3.3.3.4.1	4.3.2.3	X		1	X			
Melting point	3.3.3.4.1	4.3.2.6	X		1	-	Percent of delamination		
Basis weight	3.3.3.4.1	4.3.2.7		X	1	X	Ounce		
Polyethylene:									
Thickness	3.3.3.3	4.3.2.4		X	Avg. of 3	-	0.0001 inch		
Odor:									
(when applicable)	3.3.9	4.3.2.5	X			X			

1/ If failure is indicated, report either description or numerical point of failure, as applicable.

2/ Test report shall include all values on which results are based.

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4.3 Tests.

4.3.1 Tests for paper. The paper test specimens shall be conditioned and tested in accordance with TAPPI standards as indicated in table XIX.

TABLE XIX. Tests for paper

Tests	TAPPI No.
Atmospheric conditions	T402
Tensile breaking strength (dry)	T404
Basis weight	T410
Internal tearing resistance	T414
Sizing	T441
Tensile breaking strength (wet)	T456
Stretch of paper and paperboard	T457

4.3.2 Tests for laminated textile bags (type III).

4.3.2.1 Bonding agent tests. The laminated adhesive shall be tested by cutting a strip 6 inches long from the cemented seam and a sample 8 inches by 8 inches of the laminated material, and immersing for a period of 6 hours in water maintained at a temperature of 68° to 70°F. The seam or laminated bond shall show no delamination at the end of this period.

4.3.2.2 Bleeding. Test for bleeding of laminated materials in type III bags shall be in accordance with ASTM D 1327.

4.3.2.3 Cracking. Tests for cracking of laminated materials in type III bags shall be in accordance with ASTM D 1327.

4.3.2.4 Thickness of polyethylene (see 3.3.3.3). A specimen of approximately 3 by 3 inches square shall be cut from an unsewn portion of a bag and the adhered material shall be carefully removed from the polyethylene before calipering. The gage used for measurement of thickness shall be of a dead weight type equipped with a dial graduated to read directly to 0.0001 inch. The area of the presser foot and the weight of the parts connected therewith shall be such as to apply a pressure of 2.00 $\pm$ 0.05 pounds per square inch to the specimen. The presser foot and the anvil surface shall be plane to within 0.0001 inch and parallel to each other within 0.0001 inch.

4.3.2.5 Odor. Tests for odor in materials used in construction of bags for subsistance items shall be in accordance with FED-STD-101, method 4018.

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4.3.2.6 Melting point of asphalt bonding agent (see 3.3.3.4.1). A specimen 4 by 4 inch square shall be cut from an unsewn portion of the bag and placed in an oven at a temperature of 140°F. (± 5°F.) for a period of not less than 2 hours. After the heating cycle, remove specimen from the oven and inspect the specimen for any delamination. Percent of delamination shall not exceed 5 percent of the bonded area of specimen tested.

4.3.2.7 Basis weight of asphalt. Cut specimens with a total area of not less than 16 square inches. Obtain total weight of samples after conditioning in an atmosphere of 73° ± 3.5°F and 50 ± 5 percent relative humidity for one hour. Place the samples in an extraction apparatus of the Soxhlet or underwriters type and extract the asphalt with either Trichlorethylene, 1, 1, 1 Trichloroethane, or carbon Tetrachloride until the extract is practically colorless. Allow the paper to dry, recondition as specified herein and then reweigh. Calculate the ream basis weight of the combined plies of the base paper.

4.3.3 Breaking strength of cord (see 5.1.2.1). Test for breaking strength of cord shall be in accordance with FED-STD-191, method 4106.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. A uniform number of bags of one size and kind shall be packed per bale. The gross weight shall not exceed 200 pounds. The bale shall be overwrapped with textile laminated material, plain osnaburg cloth, or plain burlap cloth, and strapped as follows:

5.1.1.1 Overwrapping material. The textile laminated material shall conform to class I material construction. Plain osnaburg cloth or plain burlap cloth shall conform to the requirements of tables I and II, respectively.

5.1.1.2 Strapping. The bale shall be closed by rolling the side edges of the overwrap to form a seal which shall be held in place by the steel straps. The ends of the overwrap shall be folded to form a seal at each end of the bale which shall be held in place by a steel strap centered over the top, bottom, and ends of the bale. This strap shall be applied first. A minimum of four flat steel straps shall be applied girthwise to each bale. The two outside straps shall be placed approximately 4 inches from each end, tand the intermediate straps shall be placed equidistant from each other and the end straps. Straps shall be applied by a mechanical stretching tool and shall be stretched sufficiently tight so that the compression of the bale will be held and the straps will remain in place. The ends of the straps shall be sealed with a metal seal, designed to be either pinched or crimped. Ends of straps shall be broken off at the seal joint. The straps shall conform to QQ-S-781 and shall be not less than 1/2 inch by 0.020 inch. The straps shall be coated to prevent corrosion.

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5.1.2 Level B. A uniform number of bags of one size and kind shall be packed per bale. Unless otherwise specified, the gross weight shall not exceed approximately 500 pounds. Each bale shall be securely fastened with either cord or wire strapping as follows:

5.1.2.1 Cord. A minimum of four cords, two in each direction shall be securely fastened around the bale so as to give maximum support to the bale. The baling cord shall have a breaking strength of not less than 1.75 times the weight of the bundle when tested in accordance with 4.3.3.

5.1.2.2 Flat steel straps. A minimum of three flat steel straps shall be used to bind the bale. A strap shall be placed approximately 4 inches from each end and the third strap shall be positioned equidistant from the end straps. The straps shall conform to QQ-S-781, class I, and shall be not less than 1/2 by 0.015 inch.

5.1.3 Level C. The bags shall be packed to assure carrier acceptance and safe arrival at destination in compliance with National Motor Freight Classification or Uniform Freight Classification, as applicable.

5.2 Marking.

5.2.1 Civil agencies. In addition to any special marking required by the contract or order, each bale shall be marked in accordance with FED-STD-123.

5.2.2 Military. In addition to any special marking required by the contract or order, each bale shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The bags covered by this specification may be used for domestic or overseas shipment of supplies, pulverized, powdered, granular, or crystalline and of a hygroscopic or nonhygroscopic nature. The manner in which a commodity is packed governs to a great extent its condition on arrival at destination. Therefore, a determination of the proper minimum bag specification for a commodity to be packed, stored, or shipped should be given careful consideration. Among other conditions, principal factors to be considered are value of product, hygroscopic properties, chemical and physical characteristics, shipping conditions, number of transfers, destination, warehousing facilities, atmospheric conditions, and exposure to elements.

6.1.1 Waterproofing. The bags may be waterproofed by inserting a waterproof paper, or polyethylene liner in the textile bag or by laminating textile to paper with asphalt, latex, or other adhesive before manufacturing the bags. The textiles specified herein, when made into bags, shall meet the requirements for weights of contents up to 200 pounds. Greater weights of contents are permitted on approval by the contracting officer.

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6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Type, style, (class and grade as applicable), bag number, size and capacity (see 1.2.1, 1.2.2, tables II and III, and 3.3.8)
(c) Whether type I bags shall be new or used (see 3.2)
(d) Style of seams on type III bags (see 3.3.3)
(e) Grade of textile to be used for bag fabrication (see 3.3.3.1)
(f) When paper is to be stretchable and in what direction (see 3.3.3.2)
(g) When seams of type III bags are to be waterproof (see 3.3.6)
(h) Type of valve, when required (see 3.3.7)
(i) Bag manufacturers when certificate is required (see 3.3.1.2)
(j) When mildew proofing is required (see 3.3.13)
(k) When unbleached cotton is required (see 3.4)
(l) Liner when required (see 3.5)
(m) When bag top is other than open mouth (see 3.3.7)
(n) Applicable level of packing (see 5.1)

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APPENDIX

REQUIREMENTS FOR CLOSURES, PACKAGING, PACKING, AND MARKING OF FILLED BAGS

10. USE

10.1 Types I, II, III, and IV bags covered by this specification shall be closed after filling as hereinafter specified.

20. REQUIREMENTS FOR PACKAGING

20.1 The products shall be packaged by one of the following methods as specified:

20.1.1 Five pounds of the product shall be packaged in a cotton cloth bag.

20.1.2 Ten pounds of the product shall be packaged in a cotton cloth bag.

20.1.3 Fifty pounds of the product shall be packed in a textile bag.

30. REQUIREMENTS FOR PACKING

30.1 Types I, II, and IV bags. As specified, the product shall be packed 50 to 100 pounds in bulk, directly in the textile bag.

30.2 Type III bags. As specified, the product shall be packed either 50 to 100 pounds in bulk, directly in the type III shipping bag; or 5 pounds in a cloth bag, 12 of which shall be packed in the laminated textile shipping bag (see 30.2.1), or 10 pounds in a cloth bag, 6 of which shall be packed in the type III shipping bag (see 30.2.1); or 50 pounds in a cloth bag, 1 of which shall be packed in the laminated textile shipping bag (see 30.2.2). The type III shipping bag shall be of the type, and class of bag, and the grade of textile specified in the contract or order.

30.2.1 Twelve filled and closed type IV bags of 5 pounds capacity each, or six filled and closed type IV bags of 10 pounds capacity each, shall be packed into the type III shipping bag, one on top of the other with faces of the inner bags down and ends of the inner bags adjacent to the side edges of the type III shipping bag. The type III shipping bag shall tightly enclose the contents, leaving no space that will permit shifting of the inclosed inner bags.

30.2.2 One filled and closed type IV bag of 50 pounds capacity shall be packed into the laminated textile shipping bag with the factory closed end down and the side edges adjacent to the side edges of the outer bag.

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The outer and inner bags shall be of proper relative sizes so that after packing with the required quantity of product, sufficient space is left to allow the inner bag to conform to the shape of the outer bag during handling and stacking.

40. REQUIREMENTS FOR CLOSURES

40.1 After filling, the top closure of bags shall be made by one of the following methods.

40.1.1 Types I, II, and IV bags. Types I, II, and IV bags shall be closed after filling by sewing a line not less than 1 inch from top of the cloth and parallel to the top. Closure formed shall be sufficiently strong to develop at least 80 percent of the strength of the material when tested as specified in 4.3.2.

40.1.1.1 Types I, II, and IV bags with liners. Type I, II, or IV bags with liners shall be closed in accordance with 40.1.1.1.1 or 40.1.1.1.2 as applicable.

40.1.1.1.1 Paper bag liner. After the combined bag is filled, the paper bag liner shall have the top edges brought together and then folded down against the top of the contents with a minimum of three flat folds allowing sufficient slack for the product to shift. The bag shall be closed as specified in 40.1.1.

40.1.1.1.2 Polyethylene bag liner. After the combined bag is filled, the top of the polyethylene bag liner shall be gathered together and tied or top may be heat sealed. The bag shall be closed as specified in 40.1.1.

40.1.1.2 Types I, II, and IV bags with valves. Tuck-in-sleeve valved bags shall have the sleeves folded down and back into the valve space directly under the sleeve in a manner which will provide a securely held positive closure against sifting.

40.1.2 Type III bags. After filling, the top closure shall be made by one of the following methods.

40.1.2.1 Flat or single turnover seams. Sewn with a flat seam or a single turnover seam made in accordance with 3.3.4.4. Sewn seams shall be water-proofed as specified in 3.3.6 and the temperature of the sealing compound shall be such that it will not cause delamination of the bag material.

40.1.2.2 Sewn and taped top closure. Sewn with a flat seam and taped as specified in 3.3.5.2 for bottom closures.

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50. INSPECTION AND TEST PROCEDURES

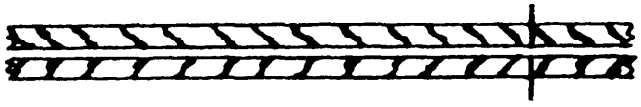
50.1 Inspection. Bags shall be inspected to determine compliance with requirements of this appendix. Sampling shall be conducted in accordance with the provisions of MIL-STD-105.

50.1.1 Examination of top closure of bag. Classification of defects shall be as indicated below. Sample unit for this examination shall be one filled and closed bag. Lot size shall be expressed in terms of bags. The sample size for examination shall be the number of sample units indicated by inspection level S-3, and the acceptable quality level (AQL) expressed in defects per 100 units shall be 4.0 for major defects and 10.0 for total defects.

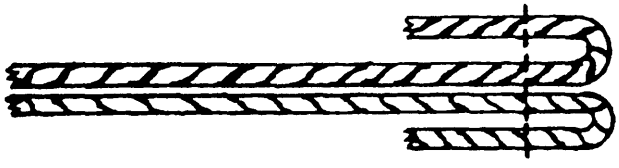
Examination for closure

Examine	Defect	Categories Major	Minor
Closure:			
Type I, II and IV bags	Not sewn closure	151	
	Stitching less than 1 inch from top of cloth		201
	False seam	152	
	Not closed	153	
Paper bag liners	Not folded as specified (see 40.1.1.1.1)	154	
Poly bag liners	Not tied or heat sealed as specified	155	
Valve top (when specified)	Not folded and closed as specified	156	
Type III bags	Not sewn with flat seam or single turnover	157	
	Seam less than 1/2 inch in width		202
	Sewn seam not taped as specified (see 40.1.2.2)	158	
	Tape not material specified	159	
	Tape less than 2-1/2 inches in width	160	
	Tape does not extend equally over and across face of bag ± 1/8 inch		203
	Sewn seams not uniformly and adequately dipped in sealing compound to a depth of not less than 1/4 inch beyond upper edge of the seam	161	

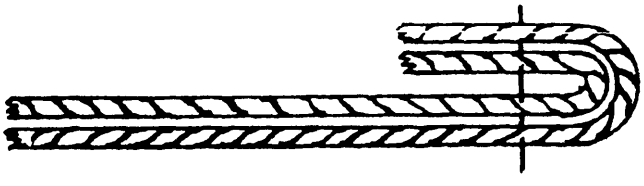
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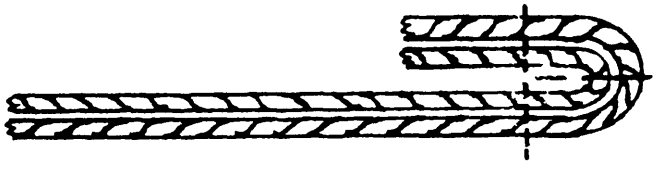
FLAT SEAM



EXPORT SEAM



SINGLE TURNOVER



CALIFORNIA SEAM



INDICATES FABRIC

INDICATES STITCHING

FIGURE 1

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MILITARY CUSTODIANS:

Army - GL
Navy - SA
Air Force - 69

Review activities:

Army - EA
Air Force - 71,80,82,84

User activities:

Army - EL,ME,WC,MD
Navy - YD,MC,SH

Civil Agency Coordinating Activities:

GSA - FSS
USDA - APS

Preparing activity:

Army - GL

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DOCUMENT IDENTIFIER AND TITLE: PPP-B-35C

Bags, Textile, Shipping, Burlap, Cotton and Waterproof Laminated

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

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